

Short Rotation Windows and an Infinite Family of Collatz Itinerary Exclusions

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Abstract

For every integer $m \geq 6$, set $t_m = (\sqrt{m^2 + 4} - m)/2$ and $\alpha_m = 1/(1 + t_m)$. We prove that no natural-number orbit of the shortcut Collatz map has the mechanical parity itinerary of slope α_m , for any real intercept, even after an arbitrary finite orbit prefix. These are distinct irrational slopes approaching one. The proof uses an elementary short-window lemma: two positive bracketing errors $0 < \varepsilon \leq \delta$ force a visit to $[1 - \delta, 1)$ among the first $q + Q$ rotation points. Maximizing a fractional part proves this directly, without a three-distance theorem. We combine this with determinant-one separation, a uniform exponential size margin, and explicit integer recurrences supplying every approximation certificate. The family theorem, irrationality, strict monotonicity, and an explicit bound on the distance to one are checked in Lean 4 using only standard foundational axioms. This extends the repository's previous single-slope result; it does not provide universal coverage of possible Collatz counterexamples.

1 The family theorem

Use the shortcut map

$$T(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ (3n + 1)/2, & n \equiv 1 \pmod{2}, \end{cases} \quad n \in \mathbb{N} = \{0, 1, 2, \dots\}.$$

For real α, ρ , its candidate mechanical itinerary is

$$w_{\alpha, \rho}(j) = \lfloor (j + 1)\alpha + \rho \rfloor - \lfloor j\alpha + \rho \rfloor. \quad (1)$$

For $0 < \alpha < 1$ these are binary letters; their limiting density of ones is α , by telescoping the floor differences.

Theorem 1.1. *Let $m \in \mathbb{N}$, $m \geq 6$, and define*

$$t_m = \frac{\sqrt{m^2 + 4} - m}{2}, \quad \alpha_m = \frac{1}{1 + t_m}. \quad (2)$$

For every $N, s \in \mathbb{N}$ and $\rho \in \mathbb{R}$, there exists $j \in \mathbb{N}$ such that

$$T^{s+j}(N) \bmod 2 \neq w_{\alpha_m, \rho}(j).$$

*Developed with Codex (OpenAI), building on the author's existing Collatz formalization. The Collatz conjecture remains open. No claim of priority in the mathematical literature is made.

The formal statements are `Metallic.no_itinerary` and `Metallic.no_eventual_itinerary`. The second follows by applying the first to $T^s(N)$; neither requires a divergence hypothesis. Below we prove every approximation and size obligation needed for this exclusion. For orientation only, rounded values are

m	6	7	8	10	20	100
α_m	0.860380	0.877151	0.890388	0.909902	0.952494	0.990100

The proof uses exact algebraic values. It covers the indicated slopes, not all slopes between them.

2 Short windows from a maximal fractional part

Write $\{x\} = x - \lfloor x \rfloor$. A *visit* at time n means

$$1 - \delta \leq \{n\alpha + \rho\} < 1.$$

The upper inequality holds automatically for a fractional part.

Lemma 2.1 (Short-window coverage). *Suppose q, Q are positive integers, $p, P \in \mathbb{Z}$, and*

$$q\alpha = p + \delta, \quad Q\alpha = P - \varepsilon, \quad 0 < \varepsilon \leq \delta. \quad (3)$$

For every real ρ there is a visit at some $0 \leq n < q + Q$. Consequently, every integer window $[a, a + q + Q)$ contains a visit.

Proof. Suppose there is no visit among these finitely many indices. Choose n maximizing $f(n) = \{n\alpha + \rho\}$. Then $f(n) < 1 - \delta$.

If $n < Q$, the index $n+q$ is still below $q+Q$ and $f(n+q) = f(n) + \delta$: adding $q\alpha = p + \delta$ changes the fractional part by δ , with no wrap since $f(n) + \delta < 1$. If $n \geq Q$, the index $n-Q$ is nonnegative and still in range, and $f(n-Q) = f(n) + \varepsilon$, again with no wrap since $f(n) + \varepsilon \leq f(n) + \delta < 1$. Both cases strictly increase the maximal value, a contradiction. Replace ρ by $a\alpha + \rho$ for the shifted-window assertion. $\square$

This argument requires neither coprimality nor a determinant condition. It also respects the closed lower and open upper endpoints of the visit interval. It proves exactly the coverage needed here, rather than a full classification of rotation gaps. In Lean the two forms are `rotation_visit_of_neighbors` and `rotation_window_of_neighbors`.

3 Separation and the Collatz level criterion

For completeness, we recall the two previously formalized inputs used with the new short window.

Lemma 3.1 (Determinant-one separation). *Under (3), suppose $Pq - pQ = 1$. If $0 < h < Q$ and $r \in \mathbb{Z}$, then $|h\alpha - r| \geq \delta$.*

Proof. Put $A = Ph - Qr$ and $B = qr - ph$. The determinant gives

$$h = Aq + BQ, \quad r = Ap + BP, \quad h\alpha - r = A\delta - B\varepsilon.$$

If $A \leq 0$, positivity of h forces $B \geq 1$, and $h < Q$ rules out $A = 0$. Thus $A \leq -1$, so the error is at most $-\delta$. If $A > 0$, then $A \geq 1$ and $h < Q$ forces $B \leq 0$, so the error is at least δ . $\square$

Proposition 3.2 (Existing endpoint criterion). *Let $\alpha \geq 0$, $2 \leq 3^\alpha$, $0 < \delta < 1$ and $Q \geq 3$. Suppose $q\alpha = p + \delta$, the separation conclusion of Lemma 3.1 holds, and every window of length G contains a visit. If*

$$3^{(q+G)\alpha+1}(3N + 3G + 1) < 2^{Q-3+G}, \quad (4)$$

then N does not have itinerary (1).

This is `sturmian_level_endpoint`, proved in the earlier development [1, 2]. Its proof combines the exact affine iterate formula, including its positive correction, with the divisibility forced by a repeated parity prefix. Here is the mechanism needed to interpret its hypotheses. A visit followed by a gap of at least Q produces a q -periodic block of $q + Q - 2$ letters. Distinct orbit values with matching parity prefixes must be far apart in ordinary integer size. The mechanical-word density bound supplies an upper estimate for those same orbit values. The endpoint reduction, valid when $2 \leq 3^\alpha$, lets one check only the largest possible starting time G . Equation (4) makes the two estimates incompatible. No unproved coverage assumption will remain when we apply this proposition.

4 A uniform family criterion

Call the data in (3) a *certified level* when $q, Q > 0$, $0 < \varepsilon \leq \delta < 1$, and $Pq - pQ = 1$. These are the fields of `SturmianNeighborLevel`.

Theorem 4.1. *Suppose $0 \leq \alpha \leq 1$ and $2 \leq 3^\alpha$. Assume there is a fixed $C \in \mathbb{N}$ such that, for arbitrarily large lower denominators q , there are certified levels satisfying*

$$6q \leq Q \leq Cq. \quad (5)$$

Then no natural number has mechanical itinerary of slope α , for any real intercept.

Proof. Write $a = 3^\alpha$, so $2 \leq a \leq 3$, and use the short window $G = q + Q$. Put $d = Q - 6q \geq 0$. Then

$$q + G = 8q + d, \quad Q + G = 13q + 2d, \quad G \leq (C + 1)q. \quad (6)$$

Since $q > 0$, we also have $Q \geq 6$, so the subtraction in (4) is ordinary integer subtraction. Multiplying that inequality by eight shows that it is equivalent to

$$24(3N + 3G + 1)a^{q+G} < 2^{Q+G}. \quad (7)$$

The constants

$$a^8 \leq 3^8 = 6561 < 8192 = 2^{13}, \quad a \leq 4$$

give

$$\begin{aligned} 24(3N + 3G + 1)a^{q+G} &\leq 24(3N + 3(C + 1)q + 1)6561^q 4^d, \\ 2^{Q+G} &= 8192^q 4^d. \end{aligned}$$

Thus it suffices that

$$24(3N + 3(C + 1)q + 1) \left(\frac{6561}{8192} \right)^q < 1. \quad (8)$$

For fixed N, C the left side tends to zero: a geometric sequence with base strictly below one dominates a linear factor. Choose a certified level beyond this threshold. Lemmas 2.1 and 3.1 then supply the coverage and separation hypotheses of Proposition 3.2, and (8) supplies its size bound. All these choices are independent of the intercept. $\square$

The constants are conservative. This theorem is an explicit sufficient criterion, not a claim that arbitrary slopes have such levels. Its Lean name is `no_itinerary_of_unbounded_neighbors`.

5 The integer recurrence discharges every certificate

Fix $m \geq 6$, and abbreviate $t = t_m$, $\alpha = \alpha_m$. Direct algebra gives

$$t > 0, \quad t^2 + mt = 1, \quad 0 < t \leq \frac{1}{2}, \quad (1+t)\alpha = 1. \quad (9)$$

Hence $2/3 \leq \alpha < 1$. In particular $3^\alpha \geq 2$, since $3^{3\alpha} \geq 9 > 8$.

Starting from $(q_0, Q_0, p_0, P_0) = (1, 1, 0, 1)$, define

$$\begin{aligned} q_{k+1} &= q_k + mQ_k, & p_{k+1} &= p_k + mP_k, \\ Q_{k+1} &= mq_{k+1} + Q_k, & P_{k+1} &= mp_{k+1} + P_k. \end{aligned} \quad (10)$$

Define the errors $\delta_k = (t^2)^k \alpha$ and $\varepsilon_k = t\delta_k$.

Lemma 5.1. *For every $k \geq 0$,*

$$q_k \alpha = p_k + \delta_k, \quad Q_k \alpha = P_k - \varepsilon_k, \quad (11)$$

$$P_k q_k - p_k Q_k = 1, \quad 0 < \varepsilon_k \leq \delta_k < 1. \quad (12)$$

Also $k+1 \leq q_k$, $Q_k \geq 1$, and

$$Q_k \leq (m+1)q_k \quad (k \geq 0), \quad mq_k \leq Q_k \quad (k \geq 1). \quad (13)$$

Proof. At level zero the error identities say $\alpha = \alpha$ and $\alpha = 1 - t\alpha$, the latter being (9). Under one recurrence step the lower error changes from δ to

$$\delta - m\varepsilon = (1 - mt)\delta = t^2\delta,$$

and the upper error changes from ε to

$$\varepsilon - m(t^2\delta) = t(1 - mt)\delta = t^3\delta.$$

These are exactly the errors at the next level. Expanding $P'q' - p'Q'$ in (10) gives $Pq - pQ$, preserving the initial determinant one. Since $0 < t < 1$ and $0 < \alpha < 1$, the error bounds follow.

The denominators stay positive and $q_{k+1} \geq q_k + 1$, giving $k+1 \leq q_k$. The identity $Q_{k+1} = mq_{k+1} + Q_k$ gives the lower ratio bound. For the upper one, $m \geq 1$ implies $Q_k \leq q_k + mQ_k = q_{k+1}$, and hence $Q_{k+1} \leq (m+1)q_{k+1}$; level zero is immediate. $\square$

For $m = 6$ the first levels are shown below. Level zero initializes the recurrence; the sixfold ratio condition is used from level one onward.

k	q_k	p_k	Q_k	P_k	$G_k = q_k + Q_k$
0	1	0	1	1	2
1	7	6	43	37	50
2	265	228	1633	1405	1898
3	10063	8658	62011	53353	72074

Proof of Theorem 1.1. For the fixed parameter m , choose $C = m + 1$. Lemma 5.1 provides certified levels with unbounded q_k and $6q_k \leq mq_k \leq Q_k \leq Cq_k$ for every $k \geq 1$. Theorem 4.1 excludes the itinerary from every natural seed and every intercept. Applying it to $T^s(N)$ excludes all finite-prefix shifts as well. $\square$

No continued-fraction theorem is needed to construct or validate these levels. The induction and the positive-root equation are the complete approximation certificates.

6 Distinct irrational slopes approaching one

Proposition 6.1. *For $m \geq 2$, α_m is irrational. The sequence $(\alpha_m)_{m \geq 0}$ is strictly increasing, and for every $m \geq 0$,*

$$0 < 1 - \alpha_m < \frac{1}{m+1}. \quad (14)$$

In particular, the excluded slopes form an infinite set with limit one.

Proof. For $m \geq 2$ the integer $m^2 + 4$ lies strictly between m^2 and $(m+1)^2$, so it is not a square. The square root of a nonsquare nonnegative integer is irrational; subtracting m , dividing by two, adding one, and taking the reciprocal preserve irrationality here. This uses mathlib’s proved nonsquare criterion, not an asserted axiom.

If $m < n$ and $t_m \leq t_n$, positivity gives $t_m^2 + mt_m < t_n^2 + nt_n$, contradicting that both sides equal one. Thus $t_n < t_m$ and $\alpha_n > \alpha_m$.

Finally $mt_m = 1 - t_m^2 < 1$, while $1 - \alpha_m = t_m\alpha_m > 0$. Therefore

$$(m+1)(1 - \alpha_m) = (m+1)t_m\alpha_m < (1 + t_m)\alpha_m = 1,$$

which proves (14). The same bound, with an arbitrarily large natural parameter, puts an excluded slope within every positive distance of one. $\square$

7 Formal verification and scope

The three new source modules are:

Module under lean/Collatz/	Role
<code>SturmianWindow.lean</code>	Short-window coverage for every intercept.
<code>SturmianFamily.lean</code>	Certified-level interface and uniform size criterion.
<code>SturmianMetallic.lean</code>	Recurrence certificates, family exclusion, and slope geometry.

The principal quantified type is

```
Collatz.Metallic.no_itinerary (m : ℕ) (hm : 6 ≤ m)
  (N : ℕ) (ρ : ℝ) : ¬ Collatz.HasItin (Collatz.Metallic.alpha m) ρ N.
```

`HasItin` asserts equality of the parity and mechanical letters at every natural time. The only assumptions in this type are the parameter range and the quantified seed and intercept; there is no remaining approximation, coverage, or asymptotic hypothesis.

The selected modules build with Lean v4.27.0-rc1. The reproducible command `python3 analysis/audit_theorems.py --build` checks the selected modules and prints exact types and

transitive axiom dependencies. The expanded audit contains 26 declarations, including the new coverage, family, and geometry results. All inspected dependencies are within `propext`, `Classical.choice`, and `Quot.sound`. The new modules contain no `sorry`, `native_decide`, or added mathematical axiom. This is ordinary Lean checking and axiom inspection; it is not independent kernel replay, a fresh build of every dependency, or a rebuild of every nested repository and large KL certificate.

8 What this leaves open

The maximal-fractional-part lemma removes a specific rotation-coverage dependency, and the recurrence extends a single excluded slope to an infinite explicit family. The family does not cover an interval of slopes. Its accumulation at one gives no continuity principle that would exclude neighboring slopes. Parameters 3, 4, 5 lie outside the conservative sixfold criterion; the earlier silver result treats the slope associated with $m = 2$ separately.

More fundamentally, no result here forces every positive-integer counterexample to have a mechanical itinerary. General nonmechanical itineraries, divergence at the critical boundary, and all nontrivial cycles remain outside this family exclusion. The next proof target is a useful extension from exact repetition to controlled weighted defects, together with an independent restriction on possible integer counterexample itineraries. Further exclusions alone do not supply that coverage bridge.

References

- [1] M. Sharpe, *An Unconditional Sturmian Exclusion for the Collatz Map at Slope $1/\sqrt{2}$* , September 2026. Repository file `paper/silver.tex`; formal sources `SturmianApprox.lean`, `SturmianEndpoint.lean`, and `SturmianSilver.lean`.
- [2] M. Sharpe, *Rung One of the Word-Complexity Ladder: a Machine-Verified Exclusion of an Automatic Collatz Itinerary by Mahler’s Method in Degree One*, August 2026. Repository file `paper/rung1.tex`; formal sources `Rung1.lean` and `Sturmian.lean`.
- [3] M. Sharpe, *Collatz at the Critical Line*, source repository, <https://github.com/msharpe248/collatz>. Current research ledger: `analysis/RESEARCH_STATUS.md`. Verification metadata: `analysis/theorem_audit.json`.